

JONATHAN KIM

(915) 474-8685 | El Paso, TX

jon.kim.mj@gmail.com | linkedin.com/in/jonathan-kim-j1k3 | github.com/jonkim13 | jonathan-kim.me

Education

The University of Texas at El Paso (UTEP)

Expected: May 2027

Bachelor of Science in Computer Science

GPA: 3.57

Relevant Coursework: Machine Learning, Database Systems, Operating Systems, Data Structures & Algorithms, Computer Organization, Software Engineering

Experience

Software Engineer

November 2025 – Present

PM Technologies LLC

El Paso, Texas

- Built a production Python/Flask + REST API app monitoring **7** networked 3D printers in real time.
- Traced **20** duplicate DB records to FAT32 filename truncation and shipped a deduplication fix.
- Designed a WAL-safe SQLite backup system with atomic snapshots and automatic retention.

Machine Learning Researcher

May 2024 – July 2024

The Ohio State University AI-EDGE Institute

Columbus, Ohio

- Fine-tuned Stable Diffusion 2 with LoRA on a custom Pokémon dataset, cutting training time **55%** (**11h**→**5h**).

Projects

BorderETA | *React Native, Expo, TypeScript, Supabase*

January 2025 – Present

- Built and shipped a cross-platform app to the App Store serving wait-time data to **300+** users.
- Engineered a Supabase prediction engine fusing CBP and traffic data, **40%** more accurate.
- Implemented geofencing to passively collect dwell-time data, powering the forecasting pipeline.

Sophia AI | *Python, FastAPI, PostgreSQL, Twilio, Stripe, Railway, Claude API*

January 2025 – Present

- Deployed an AI SMS assistant on Railway with a FastAPI REST backend parsing texts to book calendar events.
- Engineered an idempotent Twilio/Stripe webhook pipeline with encrypted tokens for reliable delivery.

ReconAIssance | *Python, FastAPI, OpenCV, YOLOv8, InsightFace, React*

October 2025 – Present

- Designed a real-time computer-vision pipeline detecting weapons and persons of interest at **95%** accuracy.
- Architected a multi-model fusion pipeline computing per-frame risk scores at **sub-200ms** latency.

Fight Coach (Hackathon 2nd Place) | *Python, MediaPipe, OpenCV, Flask*

April 2025

- Developed a computer-vision app analyzing pose sequences to flag form errors and count strikes in fight footage.

DUI Risk Radar (Hackathon 2nd Place) | *JavaScript, React, Firebase, HTML/CSS*

September 2024

- Led a team of **3** to build a real-time web app mapping DUI incidents and high-risk driving zones.

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL

Frameworks/Libraries: FastAPI, Flask, React, React Native, PyTorch, OpenCV, MediaPipe

Databases: PostgreSQL, SQLite, Supabase (SQL); Firebase (NoSQL)

Tools/Platforms: Git, Docker

Leadership / Extracurricular

Undergraduate Teaching Assistant

January 2025 – May 2025

The University of Texas at El Paso

El Paso, Texas

- Mentored **30+** students through Java lab exercises and debugging during sessions and office hours.

Secretary

January 2024 – January 2025

Free and Open-Source Club

El Paso, Texas